How-To Library

Informatica

# Implementing a Disaster Recovery Strategy for Informatica® Big Data Management 10.2 on Amazon AWS

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# Abstract

Disasters that lead to the loss of data, whether natural or human-caused events, are unfortunately inevitable. To properly protect your organization's and clients' data, a disaster recovery plan enables you to minimize data loss and business disruptions, and to restore the system to optimal performance. This article describes several options for implementing a disaster recovery plan for an Informatica® Big Data Management implementation on Amazon Web Services.

# Supported Versions

- Informatica Big Data Management 10.2

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# Overview

Implement a disaster recovery plan to minimize data loss and business disruptions, and to restore the system to optimal performance.

You can choose between two disaster recovery strategies for Informatica Big Data Management running on Hadoop clusters that are deployed on Amazon Web Services (AWS).

Choose from the following options:

- Pilot Light recovery
- Warm recovery

Implement disaster recovery in the following project phases:

Phase 1: Choose the strategy that fits your organization's requirements.

Phase 2: Implement the strategy.

Phase 3: Perform validation tests and report results.

# Phase 1: Choose the Disaster Recovery Strategy

This section gives detailed descriptions, along with advantages and disadvantages, of each of the strategies for disaster recovery.

## *Pilot Light Recovery*

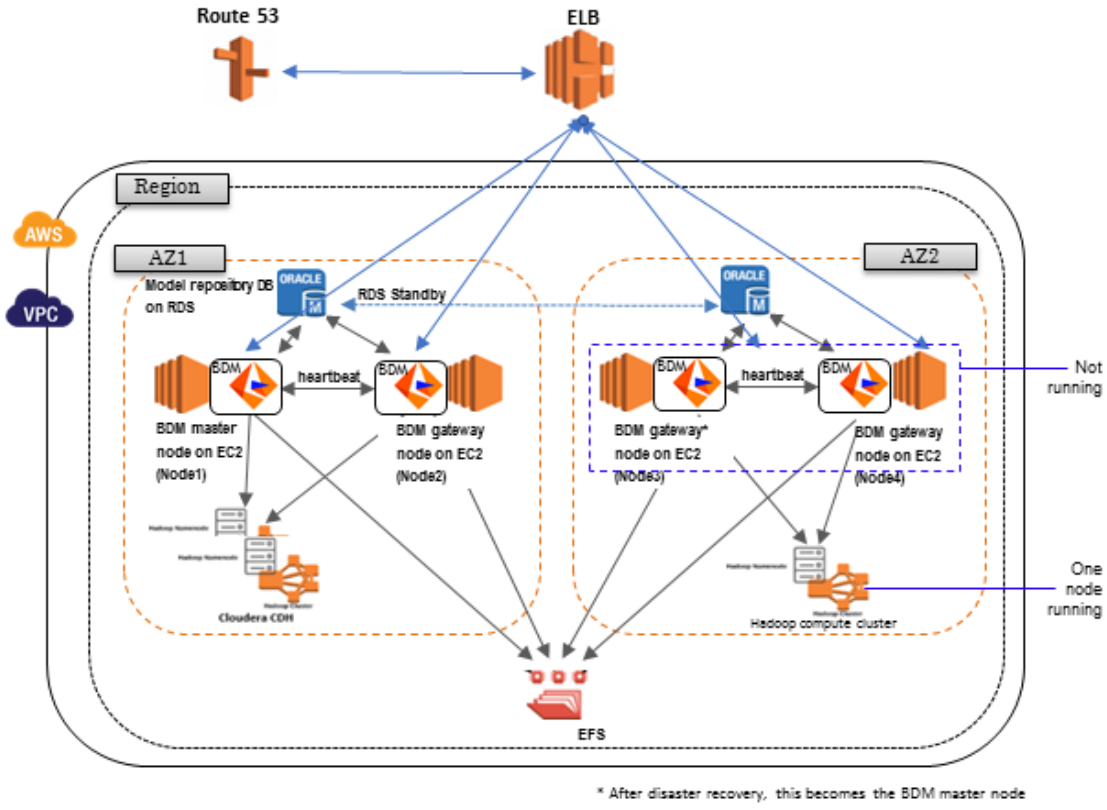
Pilot Light recovery uses Amazon RDS multi-Availability Zones (multi-AZ) to implement real-time replication of the Informatica domain, the Model repository, and the Model repository database in a standby availability zone. A system failure causes Amazon RDS to perform an automatic failover to the standby availability zone.

This strategy also uses Amazon EFS as the storage layer for applications, mappings, and workflows that the Data Integration Service runs. Amazon EFS stores data across multiple availability zones to prevent data loss from the failure of any single component.

Pilot Light recovery uses an active/passive environment. In the backup availability zone, EC2 instances are passive, and the Hadoop compute cluster is active.

The clusters in each availability zone include a Hadoop cluster that is used for big data processing. The Hadoop compute cluster includes all the replicated data from Availability Zone 1 (AZ1) Hadoop cluster nodes on a single node, where data is constantly replicated. For more information about how the Hadoop cluster handles disaster recovery, refer to the cluster vendor documentation.

The following image shows the environment for Pilot Light recovery:




Availability Zone 2 (AZ2) serves as the backup zone. Only the Hadoop cluster runs. A system failure in AZ1 causes Amazon RDS to perform an automatic failover to AZ2. The administrator then performs a series of manual steps to bring up the BDM gateway nodes and the Hadoop cluster worker nodes in AZ2, update connections, and other steps.

## Pilot Light Advantages and Disadvantages

The primary advantage of Pilot Light recovery is cost savings. Costs are minimized because AZ2 has only one running node until you start additional nodes after failure of AZ1.

The disadvantage of Pilot Light recovery is the time it takes after a failure in AZ1 to manually start nodes and services in AZ2. This downtime could affect Service Level Agreement metrics.

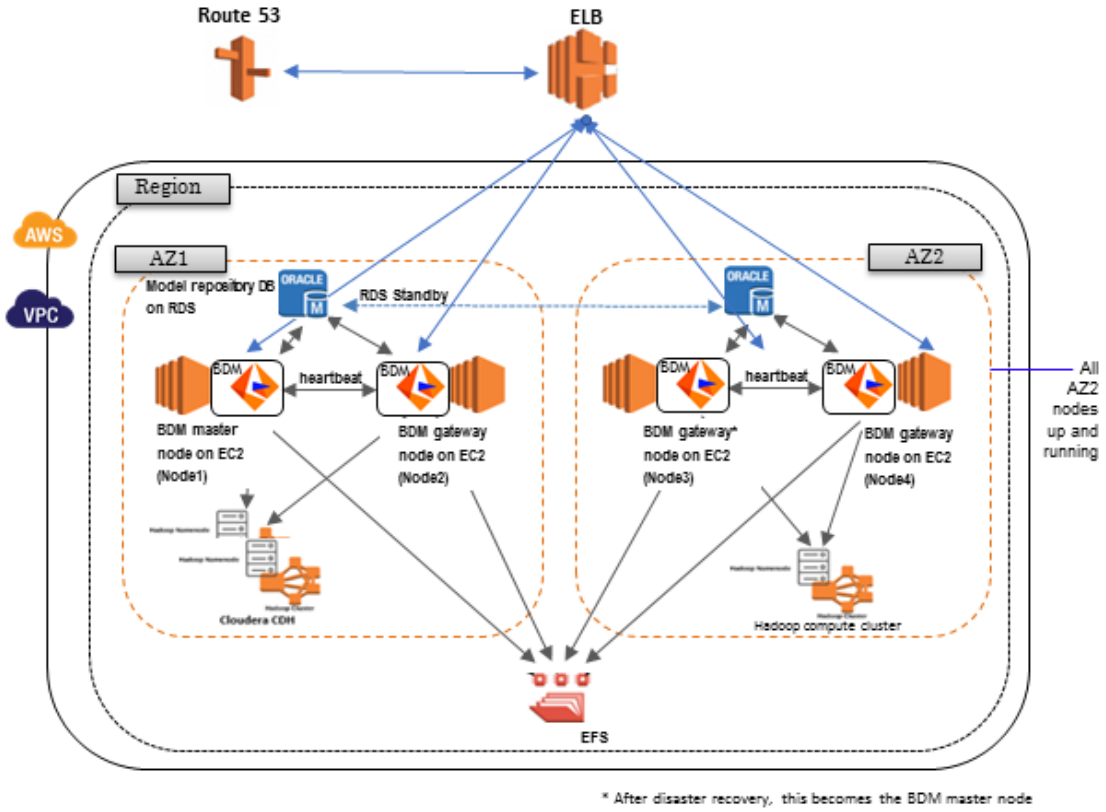
# *Warm Recovery*

Warm recovery uses Amazon RDS multi-AZ to maintain two Informatica domains in separate availability zones. Amazon RDS stores metadata for the domain, the monitoring Model repository, and the Model repository. Configure the Data Integration Service to store data on a mounted Amazon EFS volume that stores data across multiple availability zones.

Warm recovery uses an active/active environment. In the backup cluster, EC2 instances are active, and the Hadoop compute cluster is also active. In a system failure, Amazon RDS performs an automatic failover to already running instances.

The clusters in each availability zone include a Hadoop cluster that is used for big data processing. The cluster in Availability Zone 2 (AZ2) replicates the data from Availability Zone 1 (AZ1). For more information about how the Hadoop cluster handles disaster recovery, refer to the cluster vendor documentation.

The following image shows the architecture for the Warm recovery strategy:




In Warm recovery, the system in AZ1 is replicated in AZ2, where all nodes are constantly up and running.

In a failure in AZ1, Amazon RDS automatically fails over to the backup zone. Informatica gateway nodes in AZ2 become the master nodes. The domain Model repository starts automatically.

### Warm Advantages and Disadvantages

The advantage of Warm recovery is automation. All services in AZ2 start automatically so that you can start running jobs with minimal downtime.

The disadvantage of the Warm recovery strategy is the cost of maintaining a constantly running backup availability zone.

# Phase 2: Implement the Disaster Recovery Strategy

This section tells how to implement each type of disaster recovery plan.

## *Pilot Light Recovery Implementation*

Perform the tasks in this section to implement the Pilot Light recovery option.

### Verify Prerequisites

Verify the following prerequisites before you begin to set up Pilot Light disaster recovery:

#### Domain Prerequisites

- Install an Informatica domain in both zones.
- The Informatica domain stores metadata on Amazon RDS configured for multi-AZ.
- The Model repository and the Model repository database are stored on Amazon RDS configured for multi-AZ.
- The Data Information Service is configured on the Amazon EFS that stores source and target data, caches, and log files.

#### Cluster Prerequisites

- Availability Zones 1 and 2 are set up in the VPC.
- Amazon Elastic Load Balancer monitors all gateway nodes in AZ1 and AZ2.
- A Route53 alias is configured to point to the Elastic Load Balancer.
- Create and maintain AMI with a copy of the Big Data Management binary package. Apply patches on a regular basis. Store and update AMI in S3 storage to ensure the image is updated.
- Backup nodemeta.xml.
- Create a backup scheduler for connector contents, and store backup files and snapshots in an Amazon S3 bucket.
- A Hadoop cluster in installed in each availability zone.
- Define backup policies for Informatica embedded Hadoop components, such as HDFS, Hive, and HBase. Create snapshots regularly and store the snapshots in an Amazon S3 bucket.
- Prepare a validation test suite.

### Set Up Pilot Light Recovery

1. Establish two availability zones. Bring up EC2 instances in each availability zone.
2. Create an Elastic Load Balancer in the VPC. Set up Elastic Loan Balancer entries to monitor all nodes in both availability zones. Add all the gateway nodes as targets for the load balancer using the domain port 6005.
3. Configure a Route53 alias to point to the Elastic Load Balancer.

4. Install an Informatica domain in both zones. Designate the domain in AZ1 as the master node. Designate the AZ2 nodes as gateway nodes.
5. Configure the Informatica domain in AZ2.
    a. Create a Model repository on the master node, and designate each gateway node as backup.
    b. Configure the Data Integration Service to run on a grid on all cluster nodes.
    c. Create a cluster configuration to use with each cluster.
6. Verify that the required domain services exist in each Informatica domain.

After all services are created:

1. Shut down gateway nodes in AZ2.
2. Maintain one backup cluster node with cluster worker nodes and services up and running.

### Pilot Light Disaster Recovery Procedure

Perform the following steps when AZ1 fails over to AZ2:

1. Shut down primary domain and start disaster recovery process. Keep a record on start time.
2. Bring up the Informatica gateway nodes in AZ2.
3. Start Hadoop cluster worker nodes.
4. Update settings in mappings and workflows, where necessary, to point to resources and locations in AZ2.
5. Update resource paths in the .iar parameter files of deployed applications and workflows.
6. Reroute requests from client machine to AZ2.

System failover is complete.

## *Warm Recovery Implementation*

Perform the tasks in this section to implement the Warm recovery option.

### Verify Prerequisites

Verify the following prerequisites before you begin to set up Warm disaster recovery:

#### Cluster Prerequisites

- Availability Zones 1 and 2 are set up in the VPC.
- Amazon Elastic Load Balancer monitors all gateway nodes in AZ1 and AZ2.
- A Route53 Alias is configured to point to the Elastic Load Balancer.
- Create a backup scheduler for connector contents, and store backup files and snapshots in an Amazon S3 bucket.
- Define backup policies for the embedded Hadoop components, such as HDFS, Hive, and HBase. Create snapshots regularly and store the snapshots in a bucket on Amazon S3.
- Prepare a validation test suite.

#### Domain Prerequisites

- Install an Informatica domain in both zones. Designate the AZ1 domain as the master node and the AZ2 nodes as gateway nodes. Create the Model repository on the master node, and designate each gateway node as backup. Create a Data Integration Service in grid mode on all cluster nodes.

- Configure Node3 in AZ2 as a gateway node in the primary Informatica domain. This removes the necessity of using the Data Integration Service grid to reduce job execution latency.
- Configure the Model repository database and the Model repository on Amazon RDS configured for multi-AZ.
- Configure the Data Information Service on the Amazon EFS that stores source and target data, caches, and log files.

### Set Up Warm Recovery

1. Provision and start EC2 instances in each availability zone.
2. Configure sync between the Hadoop clusters in each availability zone.
3. Create an Elastic Load Balancer in the VPC. Set up Elastic Loan Balancer entries to monitor all nodes in both availability zones.
4. Configure a Route 53 alias to point to the Elastic Load Balancer.

### Warm Disaster Recovery Procedure

Perform the following steps when AZ1 fails over to AZ2:

1. Shut down primary domain and start disaster recovery process. Keep a record on start time.
2. Update settings in mappings and workflows, where necessary, to point to resources and locations in AZ2.
3. Update resource paths in the .iar parameter files of deployed applications and workflows.
4. Reroute requests from client machine to AZ2.

System failover is complete.

## Phase 3: Perform Validation Tests

This section guides you through post-recovery testing and reporting.

### *Create and Run Validation Test Suites*

Create and run validation test suites for the activated backup cluster.

Test suites consist of a series of mappings of various types, with each mapping run on each of the supported run-time engines.

1. Create mapping test suites.
2. Run validation test suites against the active backup cluster.
3. Check job status in the monitoring console.

**Tip:** A job executed before disaster recovery should retain its original time stamps. Jobs executed after disaster recovery should have new time stamps.

### *Perform Design-Time Verification*

To validate the new cluster, perform design-time activities using the new cluster.

Use the following criteria:

- Client users should receive notifications during the disaster recovery process.

- Client users must be able to reconnect to the domain after a pilot light recovery. Do not implement design changes until the last client user save.
- Jobs that are running during disaster recovery are interrupted. The console will display a monitoring status of "unknown."
- The user must perform manual cleanup on target connector contents if a job is interrupted during the disaster recovery process.

### *Reporting*

Record end time to calculate total recovery time. Use this time as the baseline for RTO.

**Note:** The recovery point depends on backup scheduler and snapshot policies defined by the customer.

## Author

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